

THE INTERVIEW

Low-Level Design Document

AI-Powered Mock Interview & Technical Assessment Platform

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1. Introduction

1.1 Purpose

This Low-Level Design document defines the internal system design, component-level behaviour, data flow, database structure, and implementation details of THE INTERVIEW platform.

THE INTERVIEW is a comprehensive AI-powered mock interview and technical assessment platform. It uses Generative AI to provide candidates with dynamic, domain-specific interview questions, proctored assessments, and real-time performance evaluations. This document serves as the complete technical blueprint for developers, testers, and maintainers.

1.2 Scope

THE INTERVIEW platform provides the following capabilities:

- Role-based JWT authentication and authorization
- Dynamic AI-generated technical interviews using Gemini
- Domain-specific static MCQ assessments
- Automated assessment scoring and attempt tracking
- Event-driven microservices communication
- Premium subscription tier unlocking advanced features
- Payment gateway integration (Stripe)
- Admin analytics and question management
- Comprehensive evaluation results and progress tracking

1.3 Technology Stack

Layer	Technology	Details
Frontend	Angular v21.2.0	RxJS, HTML5, Component CSS
Backend	.NET Web API v10.0 (net10.0)	Entity Framework Core v10.0
Gateway	Ocelot API Gateway	Routing, JWT enforcement
Database	Microsoft SQL Server	Per-service databases
Messaging	RabbitMQ	Event-driven communication
Authentication	JWT	Symmetric key validation
AI Integration	Google Gemini API	Model: gemini-2.5-flash
Payments	Stripe API	Checkout, Webhooks

2. System Overview

THE INTERVIEW platform is built on a scalable Microservices Architecture to ensure modularity, independent deployment, and high maintainability. Event-driven communication ensures robust data synchronization across services.

#	Service	Responsibility
1	API Gateway	Single entry point, routing & JWT enforcement
2	Identity Service	Authentication, registration, user profiles, notifications
3	Assessment Service	MCQ test delivery, scoring, attempt tracking
4	Interview Service	AI question generation, answer evaluation
5	Subscription Service	Stripe payments, Saga orchestration
6	Notification Service	Background worker, cross-service event consumer

3. Module Design

3.1 API Gateway

Module	API Gateway Module
Purpose	Acts as the single entry point for all frontend requests, routing them to the appropriate backend microservices and enforcing security.
Features	- Route mapping (e.g., /gateway/identity/* → Identity Service) - Upstream and Downstream routing via ocelot.json - JWT validation before internal forwarding - Internal route masking (returns 404 for unauthorized internal endpoints)
Endpoints	Routes: /api/auth/, /api/interviews/, /api/assessments/, /api/subscriptions/, /api/subscriptions/webhook/stripe. Gateway also exposes Swagger aggregation for Identity, Interview, Assessment, and Subscription services.

3.2 Identity Service

Module	Identity Service
Purpose	Handles user registration, authentication, authorization, and profile management.
Features	- Secure password hashing using BCrypt/Argon2 - Role management: Candidate and Admin - Manages IsPremium flag via Saga pattern (RabbitMQ) - In-App Notifications tracking
Endpoints	POST /api/auth/register POST /api/auth/login POST /api/auth/forgot-password/request-otp POST /api/auth/forgot-password/reset GET /api/auth/me POST /api/auth/refresh-claims PUT /api/auth/me GET /api/auth/admin/users GET /api/auth/admin/users/{id} PUT /api/auth/admin/users/{id}/role

	<pre>PUT /api/auth/admin/users/{id}/premium PUT /api/auth/admin/users/{id}/deactivate PUT /api/auth/admin/users/{id}/reactivate PUT /api/auth/internal/users/{id}/premium GET /api/auth/notifications PUT /api/auth/notifications/read</pre>
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3.3 Assessment Service

Module	Assessment Service
Purpose	Manages static, domain-specific MCQ assessments and grading.
Features	- Evaluates answers strictly based on defined correct options - Tracks AttemptNumber per domain using auto-increment query logic - Calculates dynamic MaxScore and grading percentages - Session continuity via local caching / sessionStorage
Endpoints	<pre>POST /api/assessments/start GET /api/assessments/{id}/next-batch POST /api/assessments/warm-up POST /api/assessments/submit GET /api/assessments GET /api/assessments/{id}/result GET /api/assessments/admin/all GET /api/assessments/questions POST /api/assessments/questions PUT /api/assessments/questions/{id} DELETE /api/assessments/questions/{id}</pre>

3.4 Interview Service

Module	Interview Service
Purpose	The core engine for dynamic AI-based technical interviews.
Features	- Connects to Google Gemini API to generate tailored technical questions - Generates fallback questions structured as JSON arrays - Audio/Text based answer recording flow - Asynchronous AI-based answer evaluation (Score & Feedback) - Interview states: Pending → InProgress → Completed
Endpoints	<pre>POST /api/interviews/start POST /api/interviews/{id}/begin POST /api/interviews/warm-up POST /api/interviews/{id}/fetch-more POST /api/interviews/submit GET /api/interviews GET /api/interviews/{id} GET /api/interviews/{id}/result GET /api/interviews/admin/all</pre>

3.5 Subscription Service

Module	Subscription Service
Purpose	Manages premium tier unlocks and handles payment processing.

Features	• Integrates with Stripe Checkout Sessions • Verifies Stripe Webhook HMAC signatures • Saga Pattern: payment success → SagaState transition → SubscriptionActivatedEvent → RabbitMQ → IsPremium sync
Endpoints	POST /api/subscriptions/subscribe POST /api/subscriptions/confirm POST /api/subscriptions/webhook/stripe POST /api/subscriptions/cancel GET /api/subscriptions/my GET /api/subscriptions/my/payments GET /api/subscriptions/all GET /api/subscriptions/payments

3.6 Notification Service

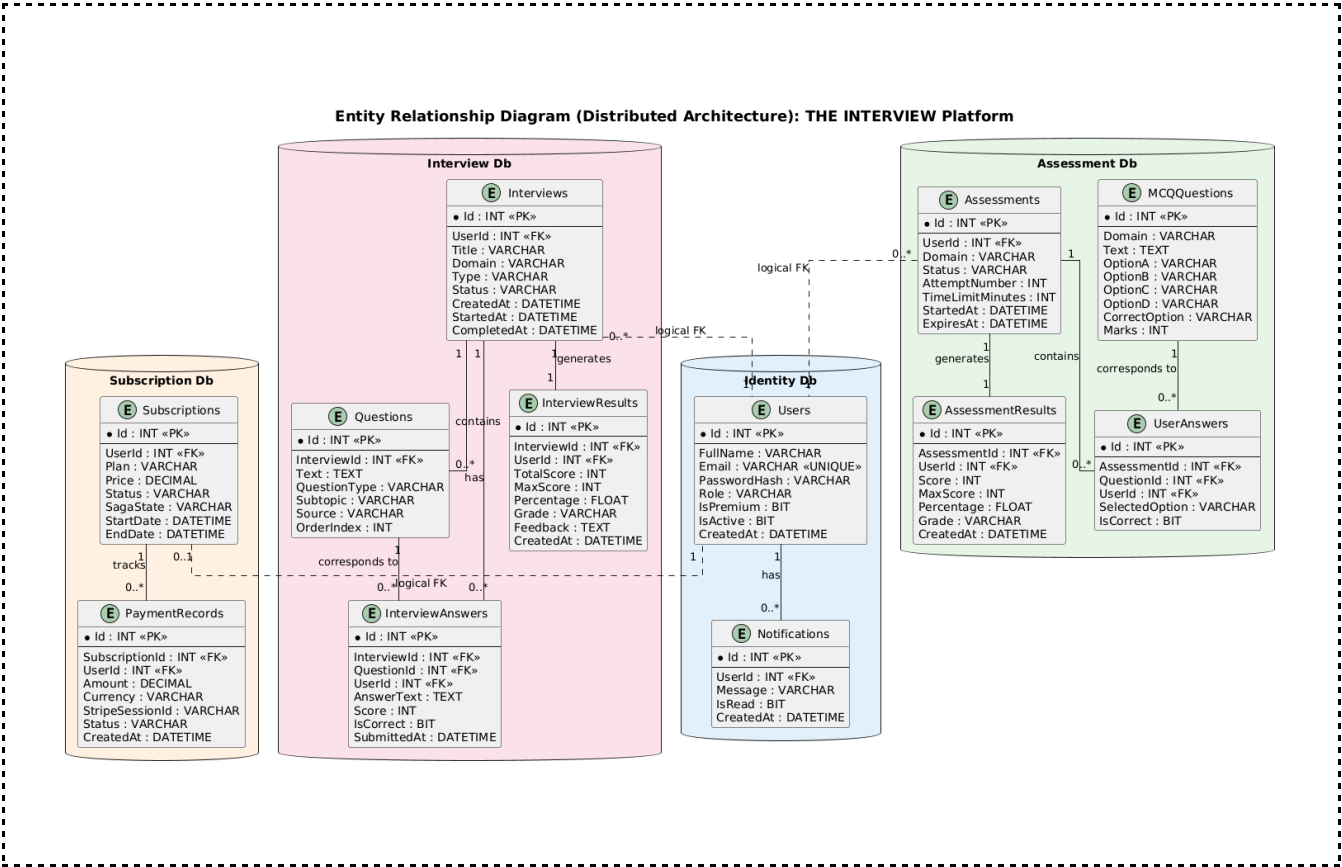
Module	Notification Service (Worker)
Purpose	Background worker processing cross-service system alerts.
Features	• Consumes RabbitMQ events (e.g., PaymentFailed, InterviewCompleted) • Calls Identity Service internally to persist notifications • High fault tolerance using Polly retry policies
Endpoints	Internal worker — no public endpoints (event-driven only)

4. Database Design

Each microservice maintains its own distinct SQL Server database instance, following the database-per-service pattern for true decoupling.

MockInterview_IdentityDb	MockInterview_AssessmentDb	MockInterview_InterviewDb	MockInterview_SubscriptionDb
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4.1 Entity Relationship Diagram



4.2 Key Tables

IdentityDb.Users

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
FullName	VARCHAR		User's full display name
Email	VARCHAR	UNIQUE	Login identifier, must be unique
PasswordHash	VARCHAR		BCrypt/Argon2 hashed password
Role	VARCHAR		Candidate or Admin
IsPremium	BIT		Toggled via RabbitMQ Saga event
CreatedAt	DATETIME		Account creation timestamp

IdentityDb.Notifications

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
UserId	INT	FK	References Users.Id
Message	VARCHAR		Notification content text
IsRead	BIT		Read/unread flag

AssessmentDb.Assessments

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
UserId	INT	FK	Cross-service reference to IdentityDb
Domain	VARCHAR		Technical domain (e.g., DSA, System Design)
AttemptNumber	INT		Auto-incremented per user per domain
Status	VARCHAR		Pending, InProgress, Completed
TimeLimitMinutes	INT		Session time limit

AssessmentDb.MCQQuestions

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
Domain	VARCHAR		Question domain
Text	VARCHAR		MCQ question text
OptionA	VARCHAR		Option A
OptionB	VARCHAR		Option B
OptionC	VARCHAR		Option C
OptionD	VARCHAR		Option D
CorrectOption	VARCHAR		Correct answer: A/B/C/D
Subtopic	VARCHAR		Topic/subtopic
Marks	INT		Marks for question
OrderIndex	INT		Ordering within domain

AssessmentDb.UserAnswers

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
AssessmentId	INT	FK	References Assessments.Id
QuestionId	INT		References MCQ question
UserId	INT		Cross-service reference to Identity user
SelectedOption	VARCHAR		User selected option
IsCorrect	BIT		Whether answer is correct

AssessmentDb.AssessmentResults

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
AssessmentId	INT	FK	References Assessments.Id
UserId	INT		Cross-service reference to Identity user
Domain	VARCHAR		Assessment domain
Score	INT		User score
MaxScore	INT		Maximum possible score
Percentage	FLOAT		Score percentage
Grade	VARCHAR		Calculated grade
IsPremiumResult	BIT		Whether detailed premium result is enabled
CreatedAt	DATETIME		Result creation timestamp

InterviewDb.Interviews

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
UserId	INT		Cross-service reference to Identity user
Title	VARCHAR		Interview session title
Domain	VARCHAR		Interview domain/configuration
Type	VARCHAR		Normal or Premium
Status	VARCHAR		Pending, InProgress, Completed
CreatedAt	DATETIME		Record creation timestamp
StartedAt	DATETIME		Interview start timestamp
CompletedAt	DATETIME		Interview completion timestamp

InterviewDb.Questions

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
InterviewId	INT	FK	References Interviews.Id
Text	VARCHAR		Question text
QuestionType	VARCHAR		Subjective/MCQ type
OptionA	VARCHAR		Optional MCQ option A
OptionB	VARCHAR		Optional MCQ option B
OptionC	VARCHAR		Optional MCQ option C
OptionD	VARCHAR		Optional MCQ option D
CorrectAnswer	VARCHAR		Optional correct answer

Column	Type	Key	Notes
Subtopic	VARCHAR		Question subtopic
Source	VARCHAR		AI, Gemini_AI, Gemini_JIT, etc.
OrderIndex	INT		Question order

InterviewDb.GlobalInterviewQuestions

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
Domain	VARCHAR		Question domain
Difficulty	VARCHAR		Easy, Medium, Hard
Text	VARCHAR		Question text
QuestionType	VARCHAR		Subjective/MCQ type
IdealAnswer	VARCHAR		Expected/ideal answer
Subtopic	VARCHAR		Question subtopic
Source	VARCHAR		Question source
CreatedAt	DATETIME		Creation timestamp

InterviewDb.InterviewAnswers

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
InterviewId	INT	FK	References Interviews.Id
QuestionId	INT	FK	References Questions.Id
UserId	INT		Cross-service reference to Identity user
AnswerText	TEXT		Candidate answer
IsCorrect	BIT		Optional correctness flag
Score	INT		AI-evaluated score
SubmittedAt	DATETIME		Answer submission timestamp

InterviewDb.InterviewResults

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
InterviewId	INT	FK	References Interviews.Id
UserId	INT		Cross-service reference to Identity user
TotalScore	INT		Total achieved score
MaxScore	INT		Maximum possible score
Percentage	FLOAT		Score percentage
Grade	VARCHAR		Calculated grade

Column	Type	Key	Notes
Feedback	VARCHAR		AI-generated feedback
IsPremiumResult	BIT		Whether detailed premium result is enabled
CreatedAt	DATETIME		Result creation timestamp

SubscriptionDb.Subscriptions

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
UserId	INT	FK	Cross-service reference to IdentityDb
Plan	VARCHAR		Subscription plan name
Status	VARCHAR		Active, Expired, Cancelled
SagaState	VARCHAR		Saga orchestration state tracker

SubscriptionDb.PaymentRecords

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
SubscriptionId	INT	FK	References Subscriptions.Id
Amount	DECIMAL		Payment amount in base currency
StripeSessionId	VARCHAR		Stripe checkout session identifier
Status	VARCHAR		Pending, Succeeded, Failed

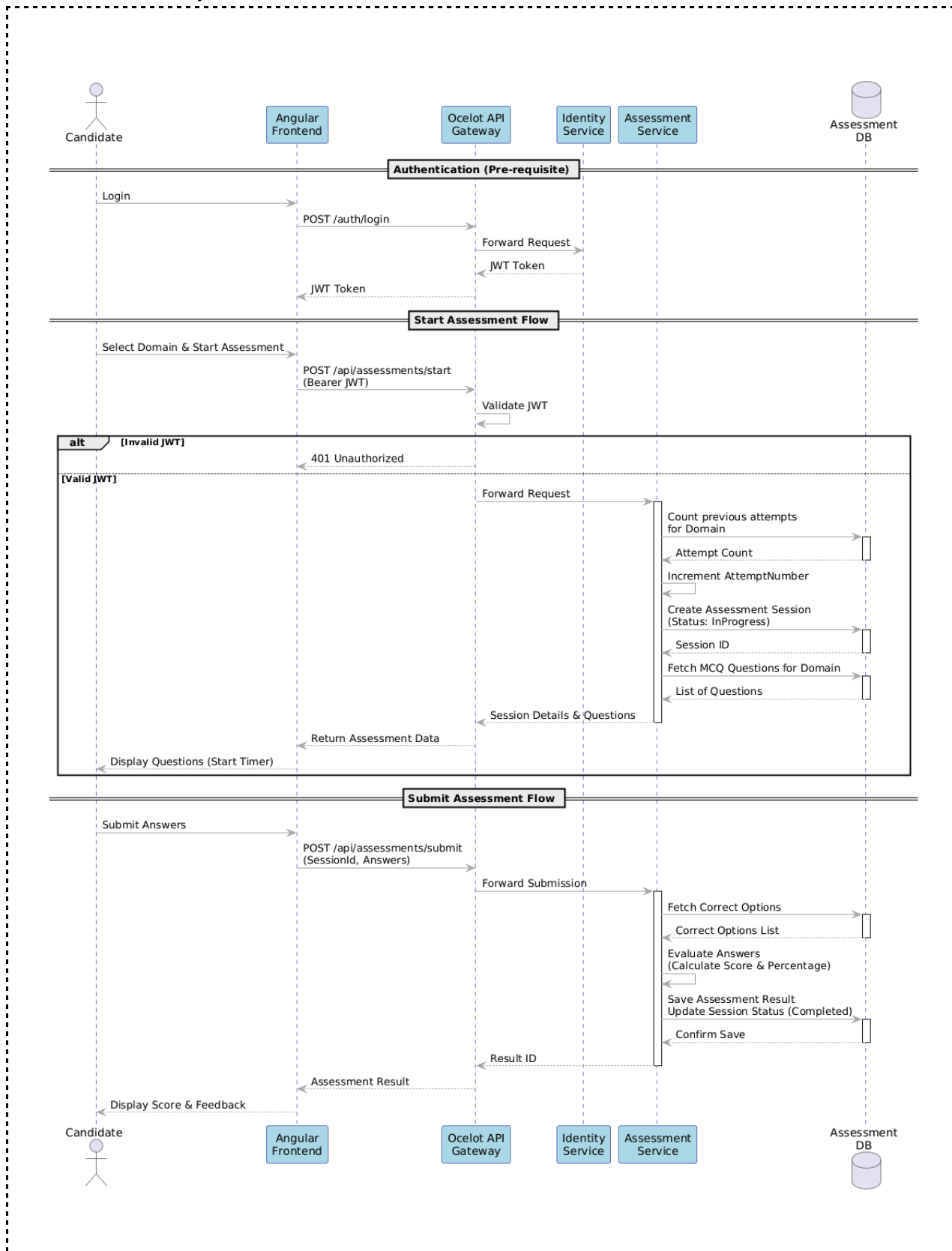
SubscriptionDb.WebhookEventLogs

Column	Type	Key	Notes
Id	INT	PK	Auto-increment primary key
EventId	VARCHAR		Stripe webhook event id
EventType	VARCHAR		Stripe event type
OrderId	VARCHAR		Stripe session/order id
PaymentId	VARCHAR		Stripe payment intent/payment id
Status	VARCHAR		Processed, Ignored, Failed
CreatedAt	DATETIME		Log creation timestamp
ProcessedAt	DATETIME		Webhook processing timestamp

5. Sequence Diagrams

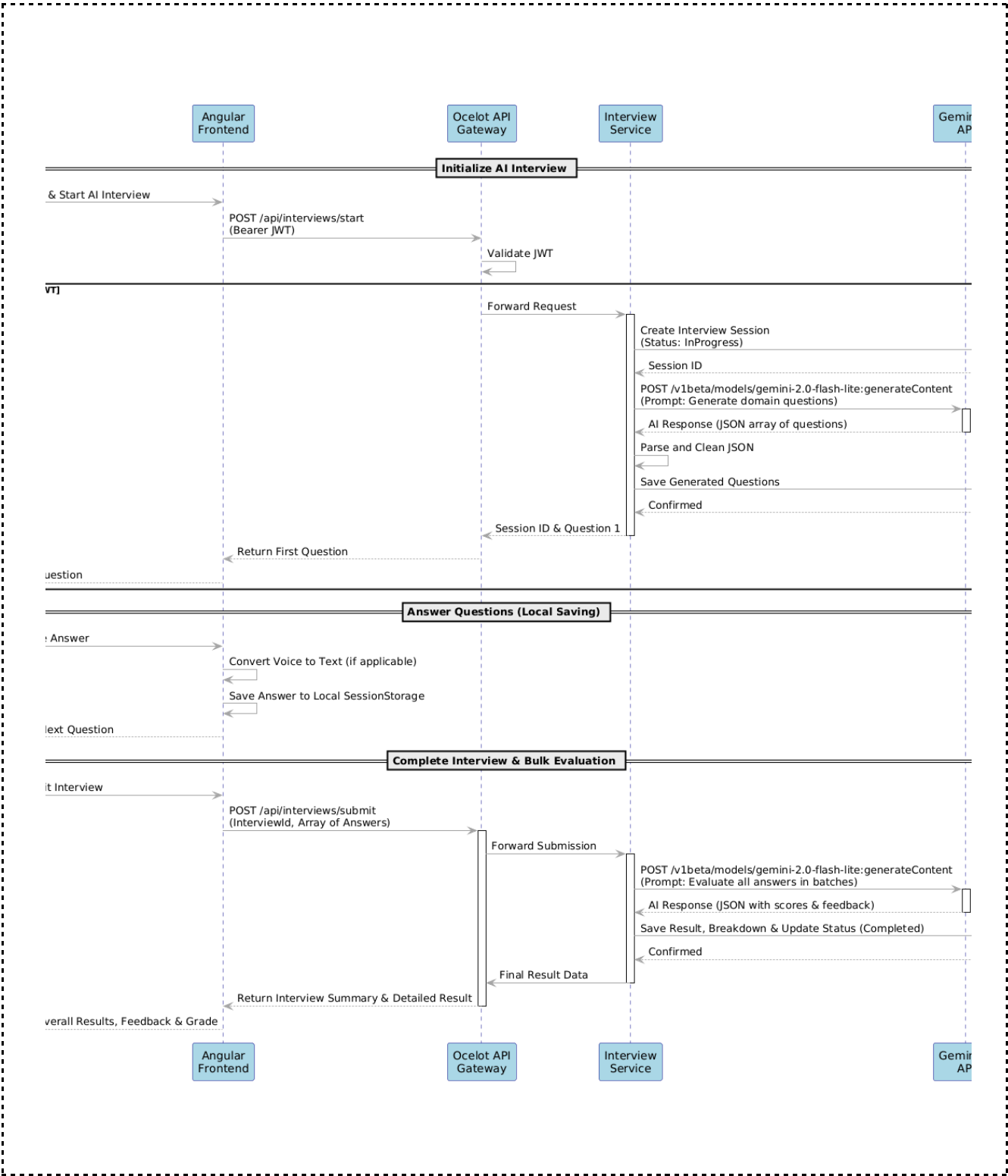
5.1 Assessment Flow

Illustrates the full lifecycle from assessment initiation to result retrieval.



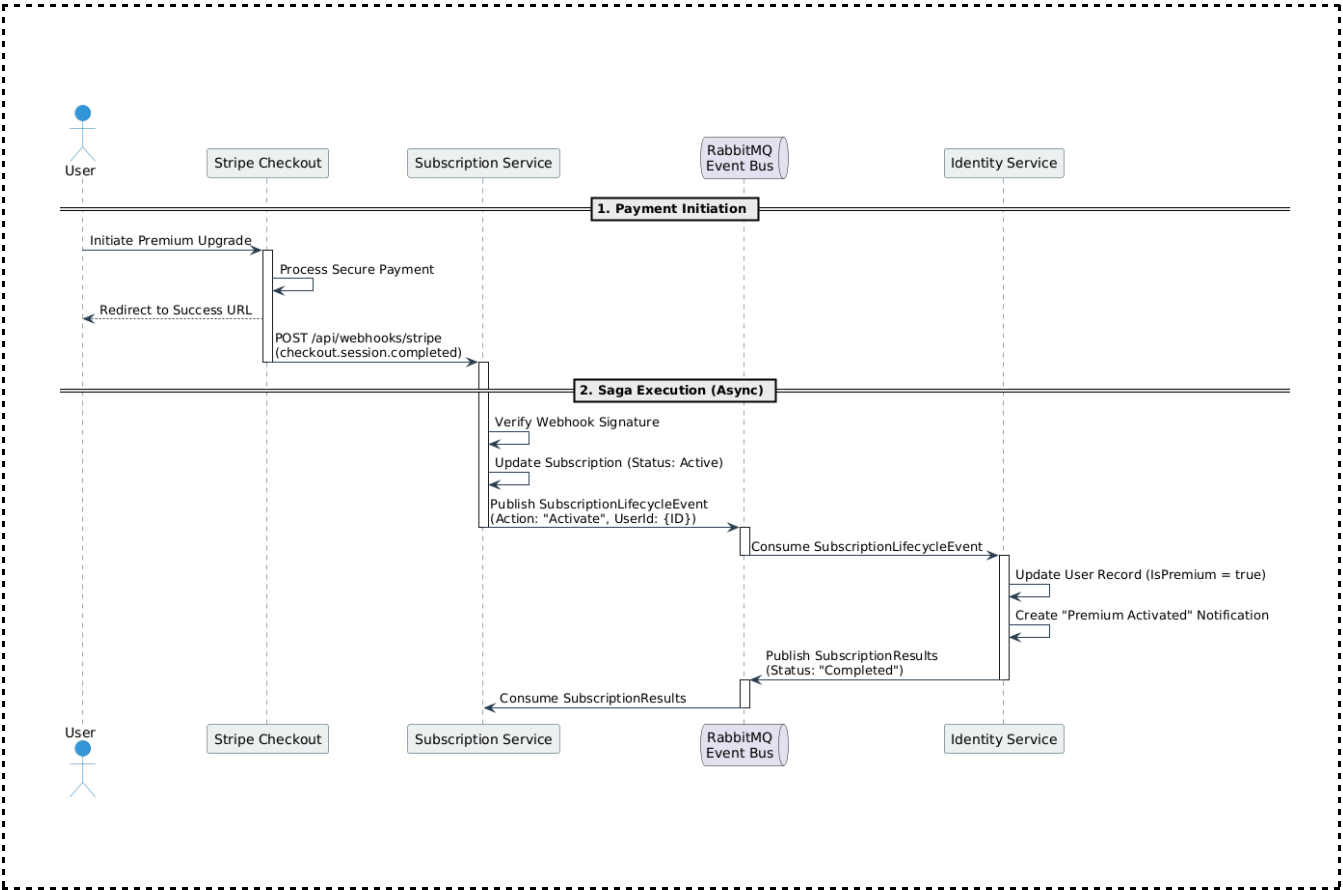
5.2 AI Interview Flow

Shows the AI-powered interview generation, answer submission, and Gemini-based evaluation pipeline.



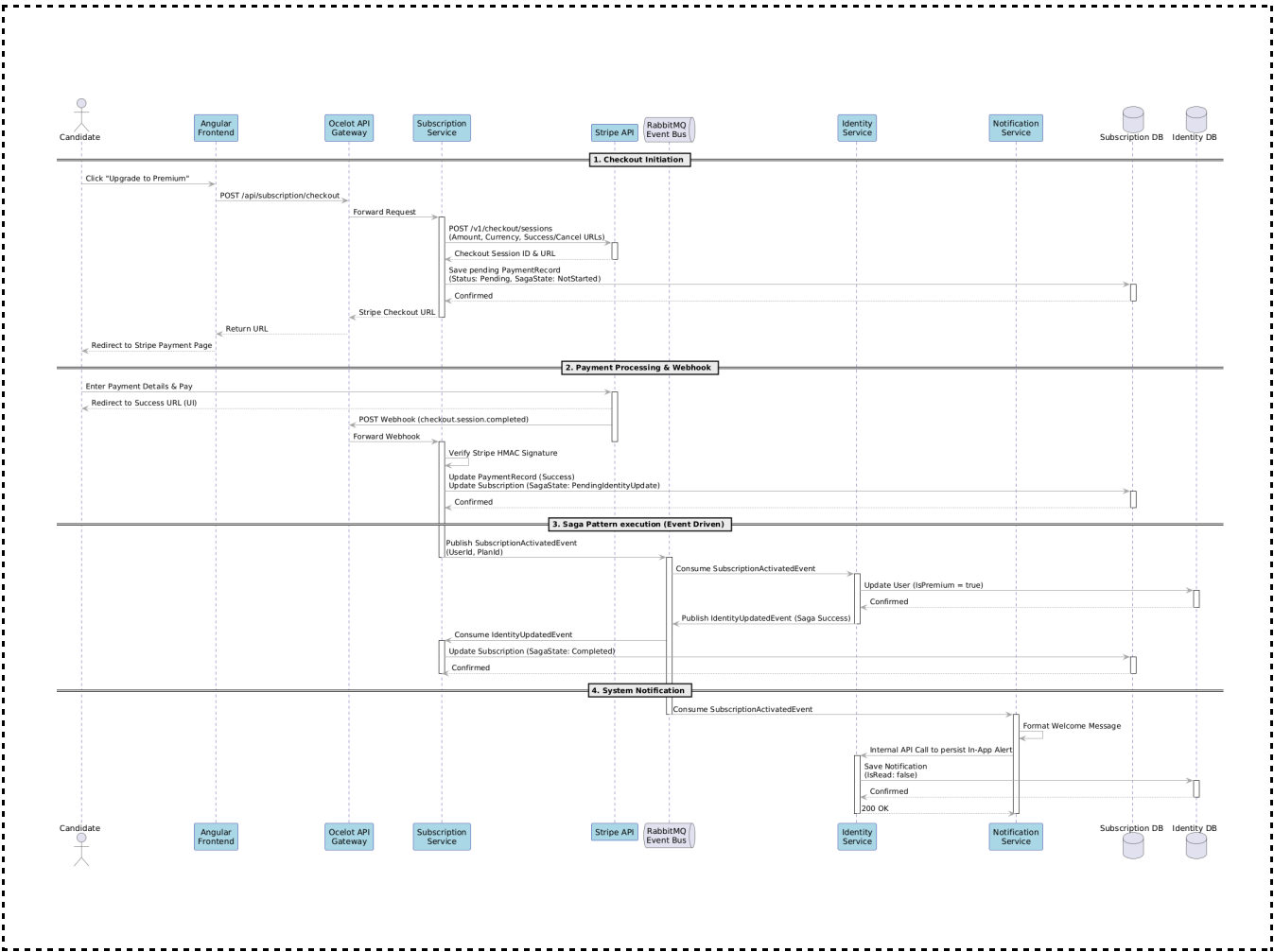
5.3 Premium Subscription (Saga) Flow

Documents the event-driven Saga pattern used to synchronize premium status across services after payment.



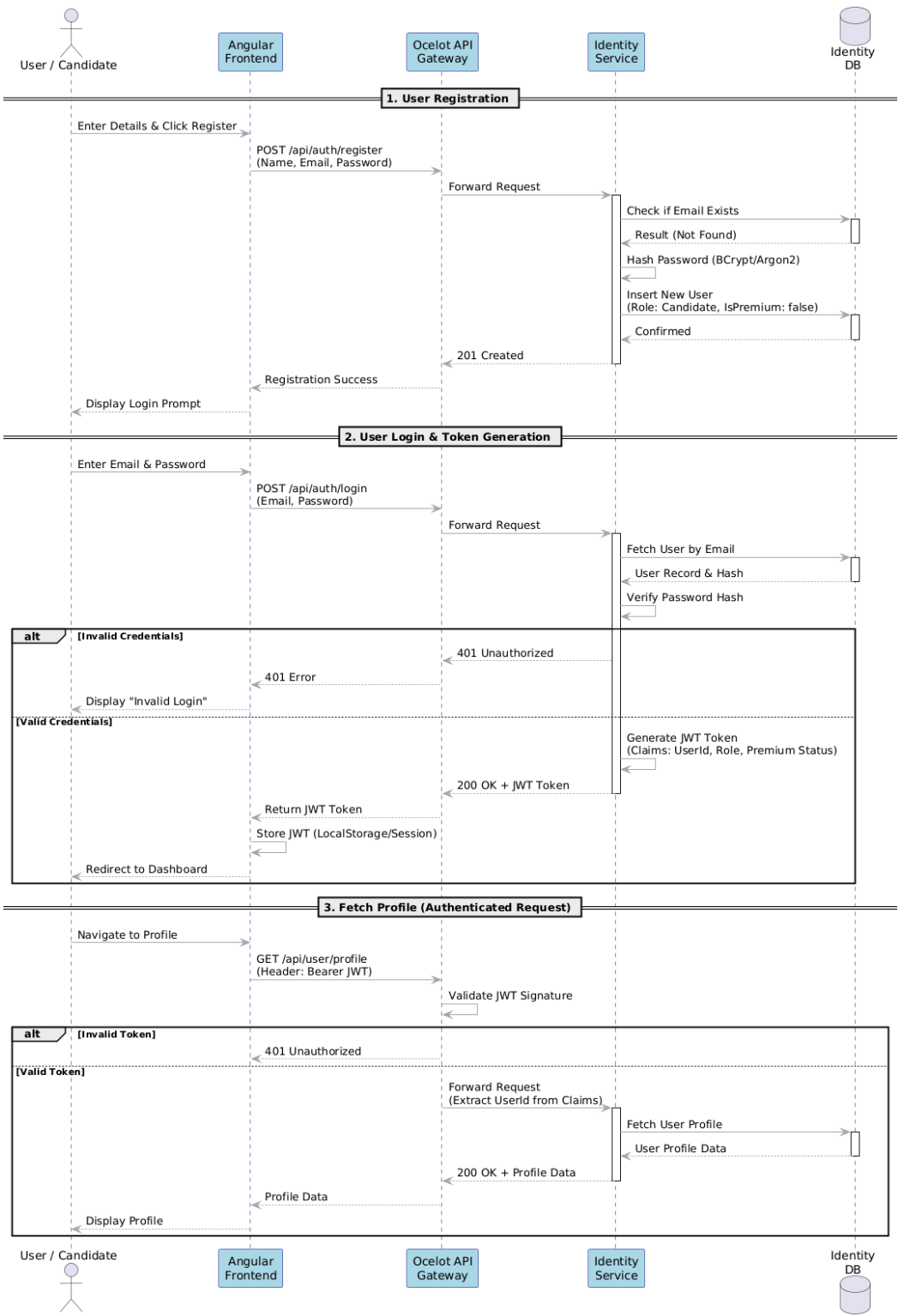
5.4 Pre-Subscription Saga Flow

Documents the pre-subscription saga state machine before payment confirmation.



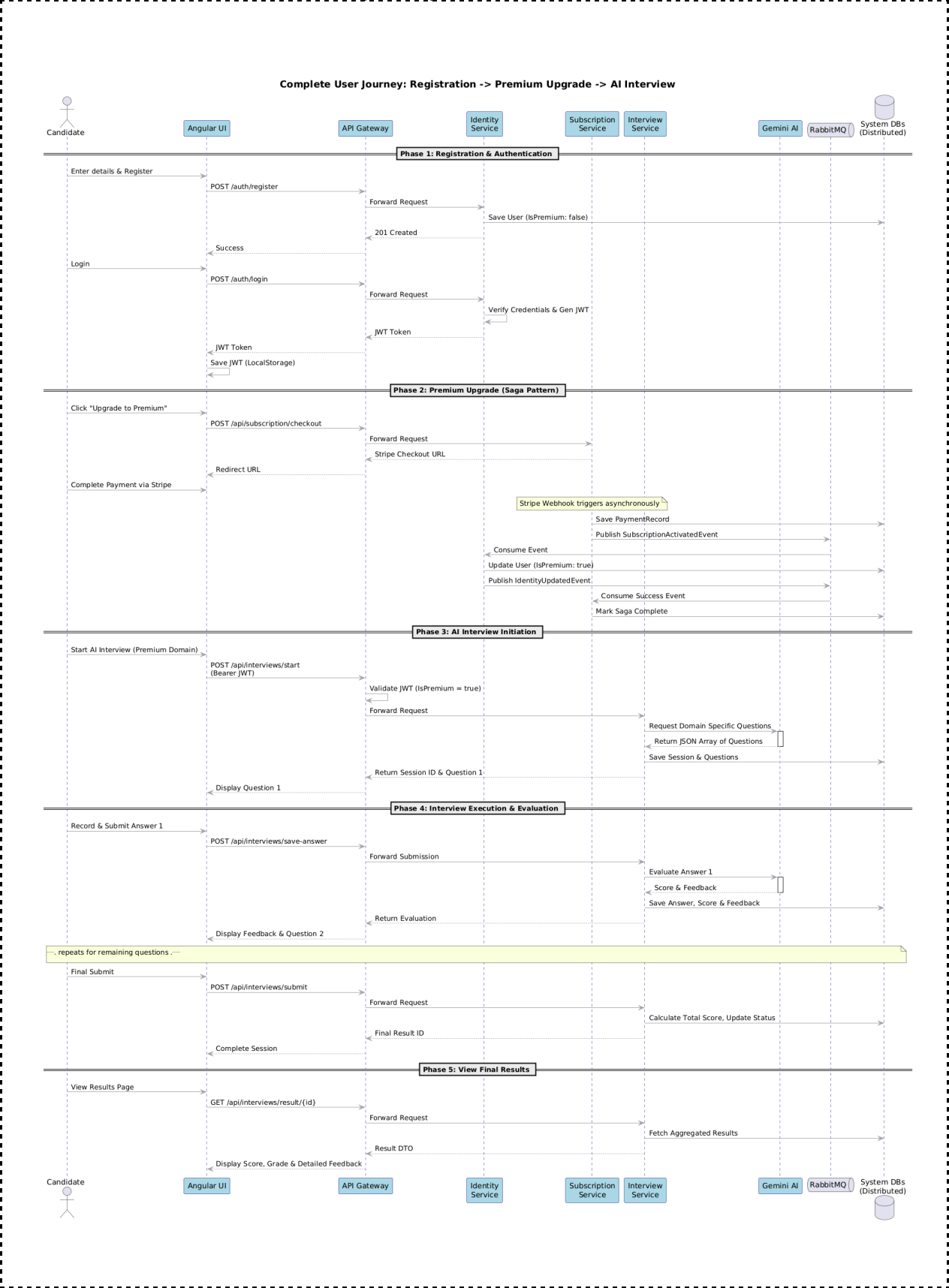
5.5 Identity Flow

Illustrates the identity and authentication flow from registration to authorized access.



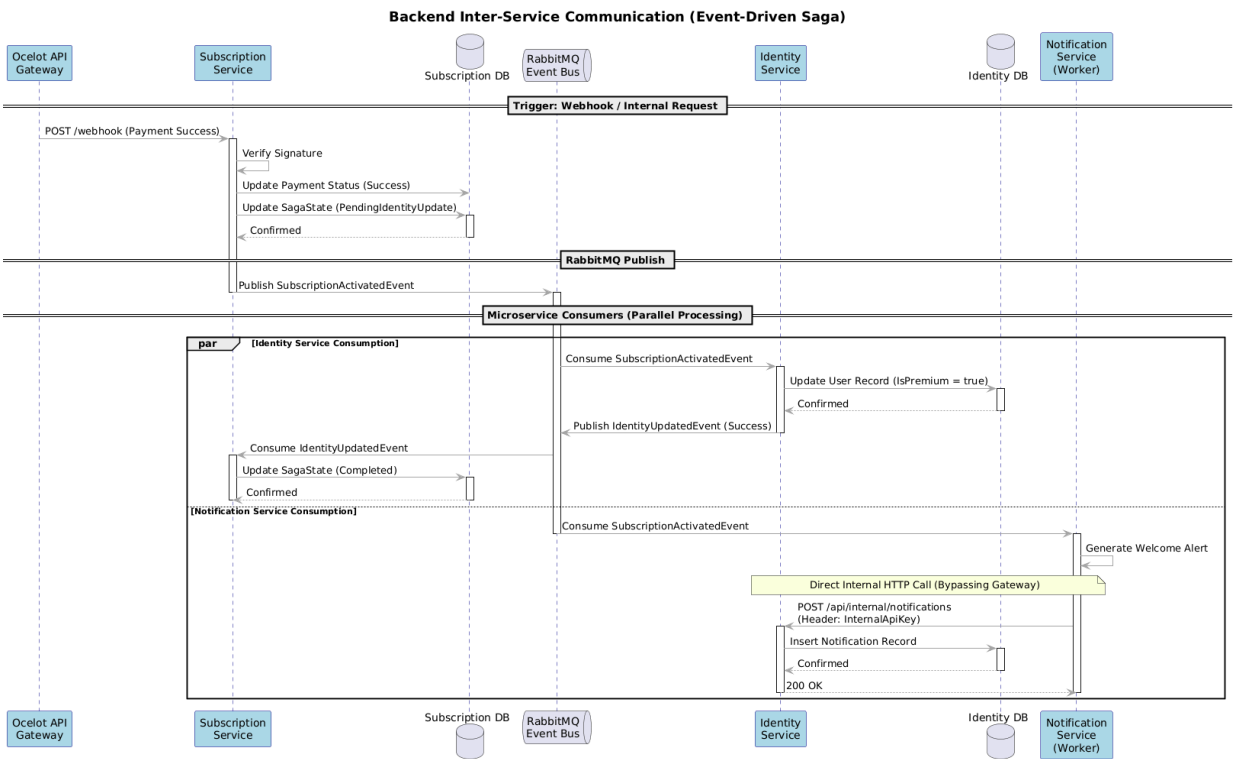
5.6 Complete System Flow

Shows the full cross-service flow covering identity, assessment, interview, and subscription interactions.



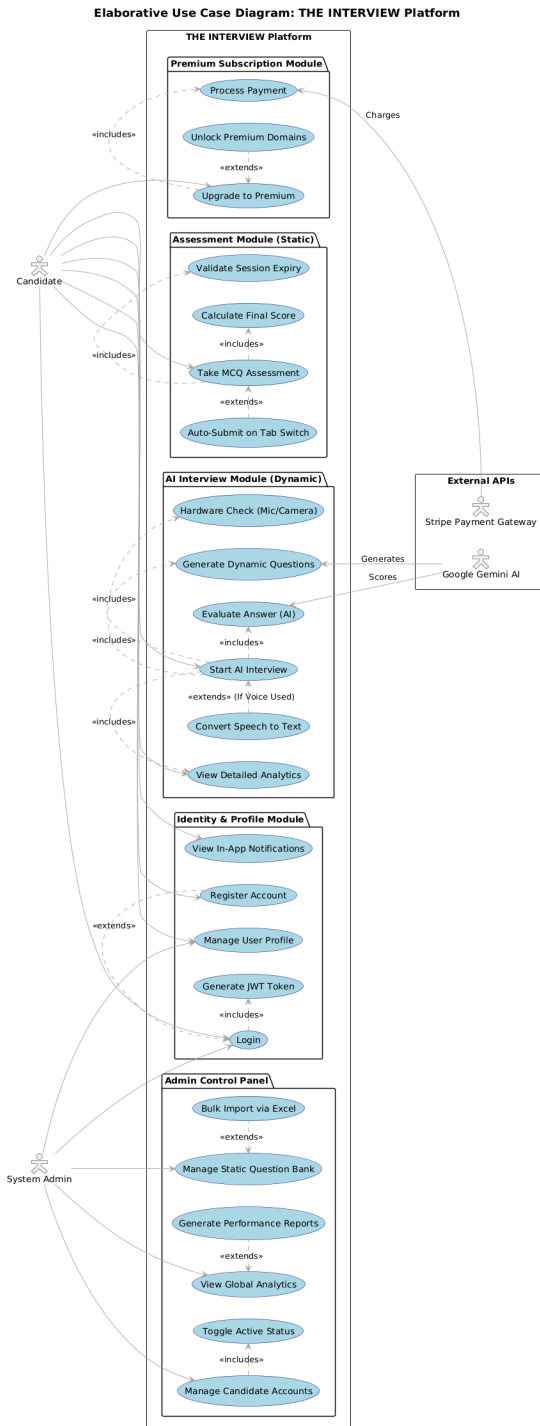
5.7 Backend Sequence Flow

Documents internal backend communication patterns across microservices including event-driven messaging.



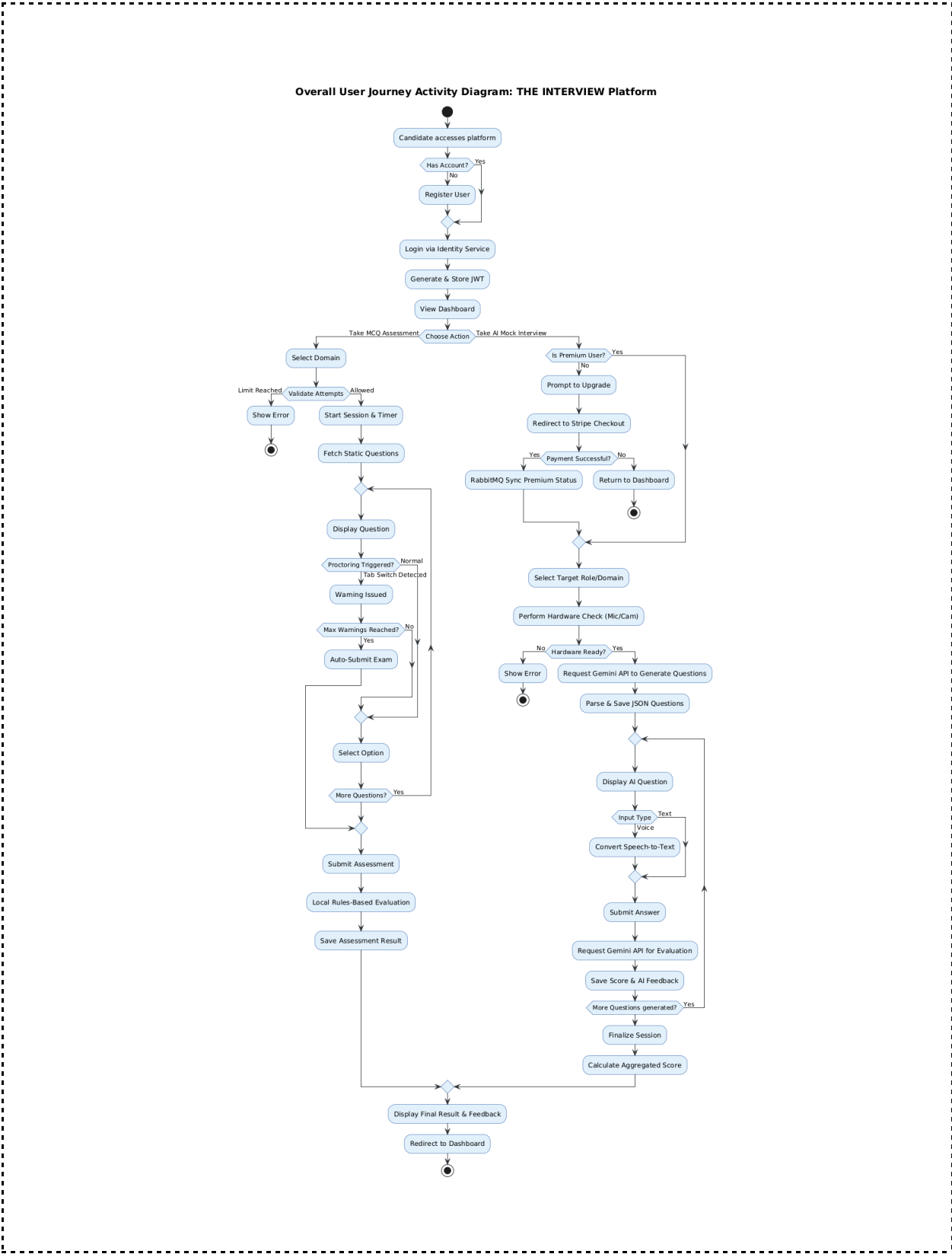
6. Use Case Diagram

Illustrates system actors and their interactions with the THE INTERVIEW platform. Actors include Candidate, Admin, and System. Key use cases span authentication, assessment, AI interview, subscription, and notification flows.



7. Activity Diagram

Depicts the step-by-step process flows for key platform operations including assessment attempts, AI interview sessions, and premium subscription activation. Shows decision points, parallel actions, and termination conditions.



8. API Reference

8.1 Identity & Profile APIs

Method	Endpoint	Description	Access
POST	/api/auth/register	Register a new user account	Public
POST	/api/auth/login	Authenticate user and issue JWT token	Public
POST	/api/auth/forgot-password/request-otp	Request password reset OTP	Public
POST	/api/auth/forgot-password/reset	Reset password using OTP	Public
GET	/api/auth/me	Fetch current authenticated user profile	User
POST	/api/auth/refresh-claims	Refresh user claims and permissions	User
PUT	/api/auth/me	Update user profile details	User
GET	/api/auth/admin/users	Fetch all registered users	Admin
GET	/api/auth/admin/users/{id}	Fetch specific user details	Admin
PUT	/api/auth/admin/users/{id}/role	Update user roles	Admin
PUT	/api/auth/admin/users/{id}/premium	Manually toggle premium status	Admin
PUT	/api/auth/admin/users/{id}/deactivate	Deactivate user account	Admin
PUT	/api/auth/admin/users/{id}/reactivate	Reactivate user account	Admin
GET	/api/auth/notifications	Fetch user in-app notifications	User
PUT	/api/auth/notifications/read	Mark all notifications as read	User
PUT	/api/internal/users/{id}/premium	Internal: Sync premium status via Saga	Internal

8.2 Assessment APIs

Method	Endpoint	Description	Access
POST	/api/assessments/start	Start assessment and calculate attempt	User
GET	/api/assessments/{id}/next-batch	Fetch next batch of questions	User
POST	/api/assessments/warm-up	Trigger warm-up question generation	User
POST	/api/assessments/submit	Submit answers for final grading	User
GET	/api/assessments	Fetch user assessment history	User
GET	/api/assessments/{id}/result	Fetch specific assessment results	User
GET	/api/assessments/admin/all	Fetch all assessments globally	Admin
GET	/api/assessments/questions	Fetch all MCQ questions	Admin
POST	/api/assessments/questions	Add new MCQ question	Admin
PUT	/api/assessments/questions/{id}	Update existing MCQ question	Admin
DELETE	/api/assessments/questions/{id}	Remove MCQ question	Admin

8.3 Interview APIs

Method	Endpoint	Description	Access
POST	/api/interviews/start	Initialize AI interview session	User
POST	/api/interviews/{id}/begin	Mark interview as started	User
POST	/api/interviews/warm-up	Generate warm-up AI questions	User
POST	/api/interviews/{id}/fetch-more	Fetch additional AI questions	User
POST	/api/interviews/submit	Final submission for AI evaluation	User
GET	/api/interviews	Fetch user interview history	User
GET	/api/interviews/{id}	Fetch specific interview details	User
GET	/api/interviews/{id}/result	Fetch evaluated interview results	User
GET	/api/interviews/admin/all	Fetch all interviews globally	Admin

8.4 Subscription & Payment APIs

Method	Endpoint	Description	Access
POST	/api/subscription/subscribe	Initialize Stripe checkout session	User
POST	/api/subscription/confirm	Confirm payment success	User
POST	/api/subscription/webhook/stripe	Process Stripe async webhooks	System
POST	/api/subscription/cancel	Cancel active subscription	User
GET	/api/subscription/my	Fetch current user subscription	User
GET	/api/subscription/my/payments	Fetch user payment history	User
GET	/api/subscription/all	Fetch all active subscriptions	Admin
GET	/api/subscription/payments	Fetch all payment records	Admin

9. Building Blocks & Shared Logic

Component	Description
ApiResponse / ApiErrorResponse	Standard success and error response contracts.
GlobalExceptionMiddleware	Converts application exceptions into consistent JSON error responses.
Custom Exceptions	AppException, ValidationAppException, NotFoundAppException, ForbiddenAppException.
RabbitMQ Publisher	IRabbitMqPublisher and RabbitMqPublisher for publishing integration events.
RabbitMQ Configuration	RabbitMqOptions and QueueNames for shared queue configuration.
Integration Events	UserRegisteredEvent, UserClaimsRefreshedEvent, SubscriptionLifecycleEvent, PaymentSucceededEvent, EmailRequestedEvent, InterviewStartedEvent, InterviewCompletedEvent, AssessmentStartedEvent, AssessmentCompletedEvent.
Service Extensions	Shared service/application extension methods for middleware and infrastructure registration.
Authorization Operation Filters	Swagger authorization helpers used in protected services.

10. Security & Compliance

Security Control	Implementation
JWT Authorization	Strict validation via symmetric keys stored in Vault/Environment Variables. All secured routes require valid tokens.
Inter-Service Security	Explicit InternalApiKey required for service-to-service calls that bypass the API Gateway.
Rate Limiting	Managed at the Ocelot API Gateway level to prevent DDoS attacks and API abuse.
Anti-Cheat Measures	Browser-level fullscreen enforcement, tab switch detection via Angular HostListeners, session invalidation on violations.
Payment Security	Stripe Webhook HMAC signature verification prevents spoofed payment events from triggering subscription activation.